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SMART OPHTHALMIC MONITORING SYSTEM USING AI FOR PREDICTIVE INTRA-OCULAR PRESSURE ASSESSMENT AND EARLY GLAUCOMA DETECTION

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ABSTRACT

Glaucoma is a major cause of irreversible blindness caused by increased intraocular pressure (IOP), Existing Tonometers are expensive, bulky, and limited to hospitals. This project presents the design of an IoT smart tonometer that enables real-time, continuous monitoring of IOP outside hospital, Uses infrared sensor to detect corneal deformation during air-puff measurement. Data processed by microcontroller and sent to cloud platform, AI model improves accuracy by removing noise and errors. Results displayed on OLED and mobile app for remote monitoring, Helps in early glaucoma detection and reduces hospital dependency.

INTRODUCTION

- Vision is important for human life and eye health monitoring is essential.
- Glaucoma occurs due to increased intraocular pressure (IOP).
- Early stages usually show no symptoms, leading to late diagnosis.
- Continuous monitoring of eye pressure is important to prevent blindness.
- Traditional Tonometers are accurate but expensive and hospital-dependent.
- Difficult for rural and elderly patients to access frequent checkups.
- IoT enables real-time monitoring and remote data access.
- AI helps analyze sensor data and improve measurement accuracy.
- Smart tonometer makes eye pressure monitoring easy and cost-effective.

PROBLEM STATEMENT

- Glaucoma often develops without early symptoms, leading to late detection and irreversible vision loss.
- Increased intra-ocular pressure (IOP) is not identified at early stages, making timely diagnosis difficult.
- Conventional devices like the Goldmann Applanation Tonometer are bulky, expensive, and require trained professionals.
- Regular eye pressure monitoring is difficult outside hospital environments.
- Patients in rural areas and elderly individuals face challenges in accessing frequent medical checkups.
- Lack of accessibility leads to irregular monitoring and delayed diagnosis.
- Traditional tonometers provide only single-time readings, without continuous tracking.

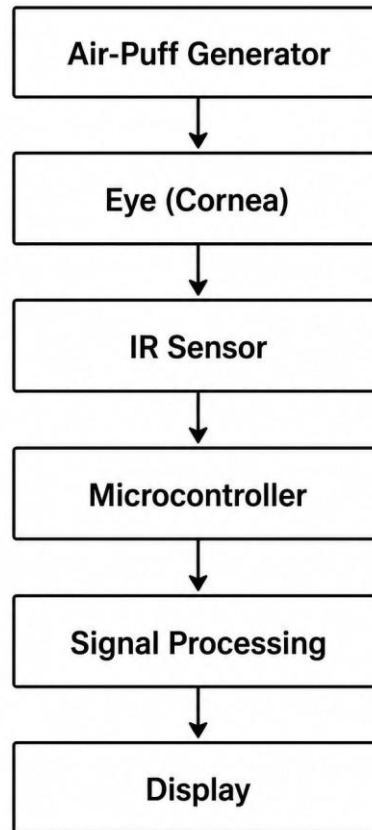
LITERATURE SURVEY

S.NO	PUBLISHER	PUBLISHED NAME	LIMITATIONS
1	Sharma A. et al. (2025), designed a portable smart tonometer for home-based glaucoma monitoring, allowing patients to measure IOP without hospital visits.	IEEE conference	Lower accuracy than clinical-grade tonometers
2	Li J. et al. (2025), reviewed wearable and implantable sensors for continuous IOP monitoring, highlighting advancements in biomedical sensing technologies.	Journal of Clinical Medicine	Invasive methods and patient discomfort issues
3	Karthik M.R. et al. (2024), developed an AI-based glaucoma detection system using deep learning models to analyze eye-related data for early diagnosis.	IEEE conference	Requires large datasets and high computational resources

EXISTING SYSTEM

- Uses non-contact air-puff technology to measure intraocular pressure (IOP)
- A controlled air pulse deforms the cornea, and the response is measured
- Infrared (IR) sensor detects corneal deformation for pressure estimation
- System is controlled by a microcontroller (Arduino-based) for data acquisition and processing
- Provides real-time IOP values through a simple display interface
- Some systems allow basic remote data sharing, but lack intelligent analysis

BLOCK DIAGRAM



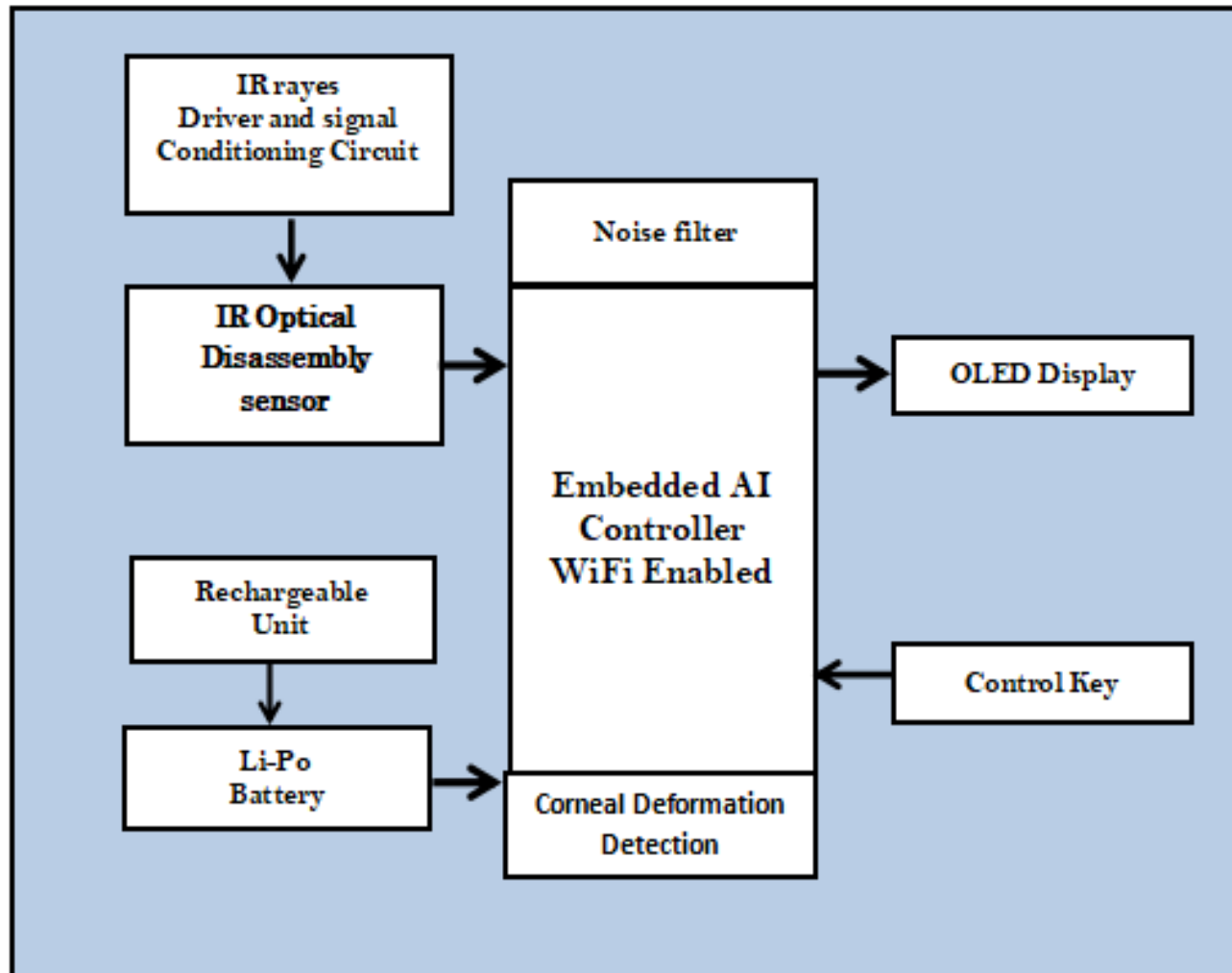
DISADVANTAGES

- Accuracy affected by eye blinking and head movement
- Sensitive to device alignment issue
- Influenced by environmental conditions
- Uses only basic signal processing
- Cannot effectively remove noise from data
- Provides limited continuous monitoring

PROPOSED SYSTEM

- IoT-enabled for home and remote monitoring of intraocular pressure. Patients can track eye health without frequent hospital visits.
- Low-cost, portable, and user-friendly design for personal use. Eliminates the need for bulky and expensive clinical devices.
- AI-assisted processing filters noise from blinking or head movement. Ensures accurate and reliable intraocular pressure measurements.
- Continuous real-time monitoring of intraocular pressure trends. Enables early detection of glaucoma for timely intervention.
- Comfortable and non-invasive measurement using air-puff or rebound probe. Minimizes discomfort while maintaining high precision and safety.

BLOCK DIAGRAM



HARDWARE REQUIREMENTS

- Infrared rays driver & Signal Conditioning Circuit
- Infrared / Optical Displacement Sensor
- Optical & Alignment Assembly
- Li-po Rechargeable Battery
- Noise Filter Module
- Embedded AI-Enabled Microcontroller (ESP32)
- OLED Display Module
- UART / Wi-Fi Interface Module

SOFTWARE REQUIREMENTS

- Programming Language:
 - Embedded C
 - python
- Integrated Development Environment (IDE):
 - Thonny IDE
 - Visual Studio

TECHNOLOGY

- Internet of Things (IoT) for real-time data transmission and remote monitoring.
- Artificial Intelligence (AI) for noise filtering and accurate intraocular pressure prediction.

Li-po Rechargeable Battery



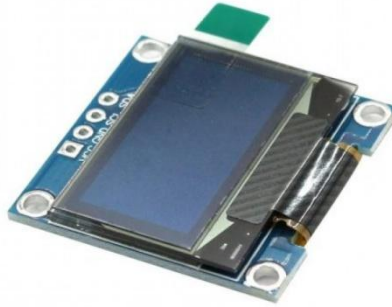
- 3.7V 300mAH (Lithium Polymer) Li-po Rechargeable Battery also known as Li-po or Lithiyam-polymer batteries are thin, light and powerful.
- This battery has a capacity of 300mAH.

ESP32 CONTROLLER



- The ESP32 is a 32-bit microcontroller developed by Espressif Systems. It acts as the main controller of the smart tonometer by collecting and processing data from the sensors.
- It features a dual-core processor with a clock speed of up to 240 MHz and has built-in Wi-Fi and Bluetooth, enabling wireless communication.
- In this project, the ESP32 receives data from the displacement sensor, processes the readings, sends them to the AI prediction model.

OLED DISPLAY



- The 0.96-inch I2C/IIC 4-pin OLED Display Module in Blue with a Green mark is a compact and versatile display solution for electronic projects and prototypes.
- With a vivid blue OLED screen measuring 0.96 inches, this module offers clear and sharp visual output.
- The inclusion of the I2C/IIC communication protocol simplifies connectivity, allowing seamless integration with various microcontrollers and development platforms.

OPTICAL-BASED DISPLACEMENT SENSORS

- Optical-based displacement sensors are precision sensing devices that measure the position or displacement of an object using light-based techniques.
- Provides high precision measurement of minute corneal movements, improving IOP accuracy.

SOFTWARE DESCRIPTION



- Thonny IDE is a simple and lightweight Python programming environment.
- Easy to use for beginners, students, and developers.
- Comes with a built-in Python interpreter (no complex setup needed).
- Used to collect and process sensor data in the smart tonometer system.
- Provides syntax highlighting, auto-completion, and error detection.
- Supports step-by-step debugging and variable tracking.
- AI algorithm improves accuracy and detects abnormal IOP values.
- Data is sent to cloud and accessed through a mobile app.
- Supports microcontrollers and IoT projects.

METHODOLOGY – IMAGE PROCESSING

- **Data Acquisition**

The system measures corneal deformation using an Infrared Displacement Sensor when a gentle air-puff or rebound probe interacts with the eye. The sensor captures displacement data related to intraocular pressure.

- **Embedded Processing**

The collected sensor signals are processed using a STM, which converts the raw analog readings into digital values and prepares the data for further analysis.

- **AI-Based Data Filtering and Prediction**

An Artificial Intelligence model processes the collected data to remove noise caused by blinking or head movement and predicts accurate intraocular pressure values.

- **IoT Communication and Cloud**

Storage Using Internet of Things connectivity, the processed IOP data is transmitted to a cloud platform where it is stored for continuous monitoring and remote access by doctors.

- **Display and Alert Mechanism**

The measured intraocular pressure values are displayed locally on an OLED Display and abnormal readings trigger alerts on a connected mobile application for early detection of Glaucoma.

ALGORITHM USED - CNN

- Convolutional neural networks are distinguished from other neural networks by their superior performance with image, speech, or audio signal inputs. They have three main types of layers, which are:
 - Convolutional layer
 - Pooling layer
 - Fully-connected (FC) layer
- The convolutional layer is the first layer of a convolutional network. While convolutional layers can be followed by additional convolutional layers or pooling layers, the fully-connected layer is the final layer.
- With each layer, the CNN increases in its complexity, identifying greater portions of the image. Earlier layers focus on simple features, such as colors and edges.
- As the image data progresses through the layers of the CNN, it starts to recognize larger elements or shapes of the object until it finally identifies the intended object.

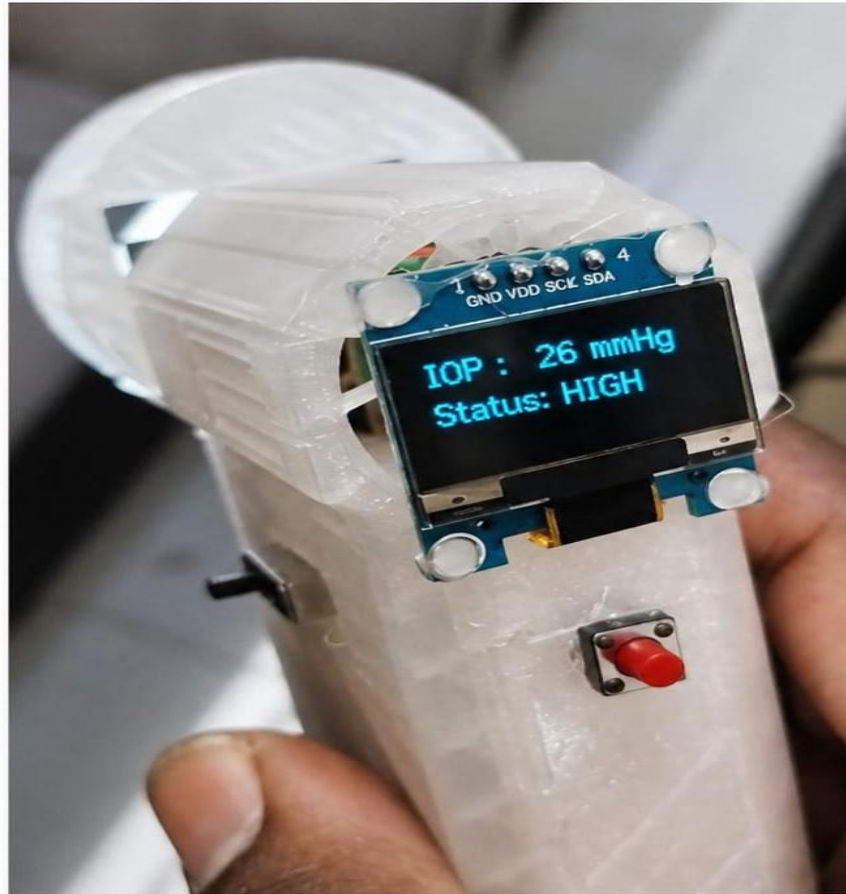
ADVANTAGES

- Enables home and remote monitoring of intraocular pressure.
- Low-cost and portable design suitable for personal use.
- AI-based noise filtering improves measurement accuracy.
- Supports continuous real-time monitoring and data tracking.
- Comfortable and non-invasive measurement improves patient convenience

APPLICATIONS

- Home-Based Glaucoma Monitoring
- Remote & Rural Healthcare Support
- Tele-Ophthalmology Integration
- Personalized Treatment Management
- Research & Clinical Studies

OUTPUT SCREENSHOT



CONCLUSION

- The Proposed IoT-based smart tonometer provides an efficient solution for monitoring intraocular pressure and supporting early detection of Glaucoma. By integrating Internet of Things connectivity and Artificial Intelligence techniques, the system enables continuous monitoring, accurate pressure prediction, and remote access to patient data.
- The portable and low-cost design allows patients to monitor their eye health at home without frequent hospital visits. In addition, real-time alerts and long-term data tracking help doctors analyze pressure trends and take timely preventive measures. Overall, the proposed system improves accessibility, accuracy, and convenience in eye pressure monitoring, helping to reduce the risk of glaucoma-related vision loss.

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